

Answer: C

$$15 \quad \frac{GM_S m}{r^2} = mr\omega^2$$

$$\frac{GM_S}{r^3} = \frac{4\pi^2}{T^2}$$

$$\therefore r^3 \propto T^2$$

$$\left(\frac{1.43 \times 10^{12}}{7.78 \times 10^{11}} \right)^3 = \left(\frac{T_{\text{Saturn}}}{11.9} \right)^2$$

$$T_{\text{Saturn}} = 29.7 \text{ years}$$

Answer: C

$$16 \quad \text{From the diagram, we can see that}$$